

BT401 Final Solved

Write two derived resources 2

2. Derived Genetic Resources

- Obsolete varieties
- Breeding lines with particular genes and performances
- Advanced cultivars
- Parents of hybrid varieties
- Cytogenetic stocks/ tester
- Mutants

What are the needs of genetic preservation 2

A **genetic preservation** is the first step in the cloning process, allowing you to produce an identical **genetic** twin or clone, which will be born at a later place in time.

A genetic preservation serves as an insurance policy for breeders and owners of valuable cattle by enabling them to extend and develop a specific bloodline when additional production is needed or untimely losses or reproductive inabilities occur.

The DNA, cryogenically preserved from the tissue sample, can permanently store the genotype of the elite donor animal, providing a genetic blueprint to recreate that cow, bull, heifer - or even a steer - at any time in the future.

Define tertiary gene pool 3

Tertiary gene pool

- Members of this gene pool are more distantly related to the members of the primary gene pool. The primary and tertiary gene pools can be intermated, but gene transfer between them is impossible without the use of "rather extreme or radical measures" such as:
- Embryo rescue (or embryo culture, a form of plant organ culture)
- Induced polyploidy (chromosome doubling)
- Bridging crosses (e.g., with members of the secondary gene pool)

What is pakistan animal resource management program 3

Pakistan is endowed with diverse **livestock genetic resources**. In fact, it is postulated that one of the centers of **animal** domestication lies in this part of the world. **Pakistan** has nearly 50 million goats. Goats are kept for milk and meat production and contribute significantly to the income of the rural farmers.

Snow Leopard	
•	Alpine Markhors

•	Asiatic Cobra
•	Mugger Crocodile
•	Himalayan Brown Bear
•	Indus River Dolphin
•	Asian Black Bear
•	Fishing Cat

What is low oxygen and low pressure storage 3

Low-pressure and Low-oxygen Storage:

For conservation of cultured plant materials low-pressure storage (LPS) and low-oxygen storage (LOS) have been developed. These are alternative methods of cryopreservation and cold storage.

What are the advantages of germplasm storage 5

Storage of seed is indispensable to most of plantation forestry, and the practice should not be dismissed too readily as a basic tool in maintaining genetic diversity. Conventional seed storage offers several advantages:

1. Seeds of many valuable species can survive long-term storage ('longterm' is defined as spanning a period of time longer than one rotation)
2. Good storage facilities are now available in most of the world, and they are used extensively for tree seed storage for various regeneration purposes;
3. Seed storage is a relatively cheap method for conserving a broad range of germplasm
4. Large land areas are not tied up in conservation
5. International exchange of genetic material is facilitated by seed storage.

What are the approaches for preservation 5

The debates regarding the best methods of digital preservation have been underway for many years (Verdegem 2006: 8). Various theoretical solutions have been proposed but each has serious flaws, such as:

- Setting of standards;
- Preserving the original technology used to create or store the records;

Migrating the software necessary to retrieve, deliver and use the records;

- Preserving records in governmental bodies;
- Conversion; and
- Emulating the original technology on new platforms.

What are the objective of research institutes for livestock

Major development objectives

The primary objectives⁵ of the Roundtable were to:

- provide a forum for senior livestock scientists and developers for the exchange of views and experiences; and
- raise the level of awareness of a far wider and influential audience with regard to the potential and the constraints facing animal agriculture in low income countries.

⁵ This sections draws on the initial presentation of Simeon Ehui, who was the principal organizer of the Roundtable at ILRI

Specific immediate objectives

The immediate objectives of the Roundtable were to:

- review the contribution and potential of livestock to increase sustainable food production, and contribute to income generation in low income countries with a forward perspective to 2020 ("The Global 2020 Vision for Livestock");
- identify major social, economic, technical and institutional constraints limiting livestock's contribution to achieving food security and economic development; and
- define appropriate strategies to alleviate these constraints and propose a framework for international action to enhance animal productivity in its broadest sense.

Expected outputs

The expected outputs of the Roundtable were:

- an analysis of past and present trends in livestock productivity and consumption of livestock products which would be used in part as an input to a "2020 Vision" paper to be further developed after the meeting;
- a statement of a defined set of objectives within the time frame specified and a related description of the constraints that must be overcome for the objectives to be achieved; and
- formulation of the set of measures ("a framework for action") or strategies needed for increasing livestock productivity in low income countries and securing better management of the natural resource base from the present to the end of the second decade of the 21st century.

Mcqs easy but from the entire course

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What is phenylketonuria? 2

Phenylketonuria (PKU)

It is a human genetic condition caused by mutations to a gene coding for a particular liver enzyme. In the absence of this enzyme, an amino acid known as phenylalanine does not get converted into the next

amino acid in a biochemical pathway, and therefore too much phenylalanine passes into the blood and other tissues. Change in environment (lowering Phenylalanine consumption) can affect the phenotype of a particular trait, demonstrating a gene-environment interaction

Define threatened species? 2

Threatened species are any species (including animals, plants, fungi, etc.) which are vulnerable to endangerment in the near future. Species that are threatened are sometimes characterized by the population dynamics measure of critical dispensation, a mathematical measure of biomass related to population growth rate

State 2 quarantine principle? 2

Isolation and **quarantine** help protect the public by preventing exposure to people who have or may have a contagious disease. ... **Quarantine** separates and restricts the movement of people who were exposed to a contagious disease to see if they become sick

Modern quarantine does not have to be absolute to be effective

- Even partial or “leaky” quarantine can reduce disease spread
- Partial quarantine can be an effective supplement to vaccination

Explain habitat loss cause of extinction? 3

Habitat destruction is the process in which natural **habitat** is rendered unable to support the **species** present. In this process, the organisms that previously used the site are displaced or destroyed, reducing **biodiversity**. ... **Habitat destruction** is currently ranked as the primary **cause** of **species extinction** worldwide.

What are molecular marker? 3

In genetics, a **molecular marker** (identified as **genetic marker**) is a fragment of DNA that is associated with a certain location within the genome. **Molecular markers** are used in **molecular** biology and biotechnology to identify a particular sequence of DNA in a pool of unknown DNA.

- RFLP (or Restriction fragment length polymorphism)
- SSLP (or Simple sequence length polymorphism)
- AFLP (or Amplified fragment length polymorphism)

- RAPD (or Random amplification of polymorphic DNA)
- VNTR (or Variable number tandem repeat)

What are three common function of quarantine?3

Quarantine practices in most countries have at least three common functions.

- The first is exclusion or regulatory actions to prevent or reduce the risk of entry of exotic pathogens, pests, or parasites along artificial pathways.
- Second is the containment, suppression, or eradication of pests or pathogens that have been recently introduced.
- Third is the assisting of exporters to meet the quarantine requirements of importing countries.

What are mechanism of cryopreservation name steps? 5

Until two decades ago the genetic resources were getting depleted owing to the continuous depredation by man. It was imperative therefore that many of the elite, economically important and endangered species are preserved to make them available when needed. Many methodologies have been devised for long term preservation of material.

Cryo-preservation

Cryo is Greek word. (krayos – frost). It literally means preservation in “frozen state.”

Cryo-preservation or **cryo-conservation** is a process where organelles, cells, tissues, extracellular

matrix, organs or any other biological constructs susceptible to damage caused by unregulated chemical kinetics are preserved by cooling to very low temperatures (typically -80 °C using solid carbon dioxide or -196 °C using liquid nitrogen)

Cryopreservation can be done at

- | | |
|---|---|
| • | |
| • | Over solid carbon dioxide (at -79 degree)
Low temperature deep freezer (at -80 degree) |
| • | In vapor phase nitrogen (at -150 degree)
In liquid nitrogen (at -196 degree) |
| • | |

60: Cryopreservation Technique:

A **cryoprotectant** is a substance used to protect biological tissue from freezing damage. Due to ice crystal formation cell membrane and cell integrity can ruptured. **cryoprotectants** are usually antifreeze

compound.

Types of Cryoprotectants:

- 1. Membrane permitting** which can freely diffuse the membrane such as **glycerol (G)**, **ethylene glycol (EG)** and **dimethyl sulfoxide (DMSO)**
- 2. Non Membrane permitting** which cannot **permeate** the cell membrane such as **sugars**

Cryopreservation technique:

•	Slow freezing and thawing
•	Rapid freezing and thawing
•	Vitrification

Ultrarapid freezing

DNA Bank. 5

DNA banks can now be considered as a means of complimentary conservation. DNA storage is particularly useful for those species that cannot be conserved in traditional seed or field genebanks and nor conserved in situ due to high risk in that area.

Advantages

DNA banking is an efficient, simple and long-term method to conserve the genetic information.

Disadvantages

There are problems with subsequent gene isolation, cloning and transfer of DNA back to a plant and it currently does not allow the regeneration of the same genotype as the original sample.

Storage strategy

Determining what to store and for how long is an important consideration, used to determine sample size, capacity of the DNA bank, preparation of samples and documentation. Long-term needs and expected volume and number of samples to be stored will determine organization and repository design.

Processing of samples

DNA preserved in DNA banks will be stored either within cells and extracted upon retrieval from storage or extracted from cells and purified before storage. The quality of the DNA is expressed through yield, purity, molecular weight, amplification efficiency and authenticity of sequences. The quality of DNA extracted from plant specimens is dependent on the condition of the specimen before storage, the storage environment and the duration of storage. Rapid drying of plant samples with silica gel or lyophilisation helps to preserve the DNA.

S torage

Once extracted DNA is a stable biomolecule, although it can easily be degraded during extraction and storage. Quality declines within days in hydrated samples held at room temperature or in refrigerators. Drying the sample or storing it in freezers or liquid nitrogen achieves better preservation of DNA molecular size. For this reason, DNA is better conserved in a form that is close to the original state and most DNA banks store cells or tissues and extract DNA upon request.

There is little information on the long-term stability of extracted DNA during frozen storage, but most repositories consider several years to decades as realistic. Information on the stability of purified DNA dissolved in buffer suggests that the overall fragment size decreases with storage time, and that the

usefulness of the specimen for PCR-based assays may be 1–2 years when stored at 4 °C, 4–7 years when stored at -18 °C and greater than 4 years when stored at -80 °C (Madisen et al. 1987; Visvikis et al. 1998). The choice of temperature usually depends on the moisture level within the sample

Difference b/w in breeding and out breeding. 5

Inbreeding:

“**Inbreeding**, the mating of individuals or organisms that are closely related through common ancestry.”

Outbreeding:

“The intentional breeding of distantly related or unrelated individuals for the purpose of producing offspring of superior quality.

Biological diversity. 2

Biodiversity, a portmanteau of biological (life) and diversity, generally refers to the variety and variability of life on Earth. According to the United Nations Environment Programme (UNEP), **biodiversity** typically measures variation at the genetic, species, and ecosystem level.

Questions

1. Define extinction?

“A species becomes extinct when the last existing member of that species dies.”

Extinction, in biology the dying out or termination of a species.

Extinction occurs when species are diminished because of ;

•	Environmental forces (habitat fragmentation, global change, overexploitation of species for human use)
•	Evolutionary changes in their members (genetic inbreeding, poor reproduction, decline in population numbers).
•	

Extinction becomes a certainty when there are no surviving individuals that are able to reproduce and create a new generation. A species may become functionally extinct when only a handful of individuals survive, which are unable to reproduce due to;

Poor health,	•
Age,	•
Sparse distribution over a large range,	•
A lack of individuals of both sexes (in sexually reproducing species)	•

3.Four main mechanism of evolution?

The allele frequency in a population's gene pool can be affected by evolutionary mechanisms, such as;

•	Mutation
•	Gene flow
•	Inbreeding Natural selection, Founder effect Random genetic drift.
•	

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Germplasm Conservation 10

In 1972, conservation of habitats rich in genetic diversity was recommended in the UN conference. Then an International Board for Plant Genetic Resource (IBPGR) was established. This board has objectives to provide necessary support for collection, conservation and utilization of plant genetic resources from anywhere in the world.

(a) In situ Conservation:

Since 1980, in situ conservation has received high priority in the world conservation strategy. The method of conservation is to preserve land races with wild relatives in which genetic diversity exists.

(b) Ex situ Conservation

It is the chief mode of conservation of genetic resources including both cultivated and wild ones. Under suitable conditions genetic resources are conserved for a long term as gene bank. Such gene bank is of two types:

- In vivo Gene Bank
- In vitro Gene Bank

(i) In vivo Gene Bank:

Generally plant seeds, vegetative propagules are used for storage for long time. The whole plants are preserved. This type of conservation strategy is called in vivo gene bank. In this approach, conservation method of storage is used for preservation of plant genetic resources

(ii) In vitro Gene Bank:

3. This approach includes the conservation of genetic resources by non-conventional methods. In this approach explants are grown on medium.

Methods of Preservation

Free Preservation or Cryopreservation:

Cryopreservation (Latin Kuos means frost) means storage of materials at very low temperature. Plant cells and tissue cultures are brought to zero state of metabolism by subjecting them to ultra-low temperature i.e. -196°C . It is done by using liquid nitrogen which provides approximately -496°C .

Cryoprotectants (e.g. glycerol, proline, mannitol, dimethylsulfoxide, sorbitol) are also used to protect the viable cells from the damage during freezing and thawing (to become unfrozen or warm). Germplasm of some plants (in the form of shoot tips, nodal or meristem explant culture) are stored at low and nonfreezing temperature ($1-9^{\circ}\text{C}$). At low temperature, growth of plant material is slow down but not completely stopped as in cryo-preservation. In cold storage there is no risk of cold injuries.

Low-pressure and Low-oxygen Storage:

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Germplasm Exploration Laboratory 5

Germplasm Exploration Laboratory:

Plant exploration is the avenue to germplasm for crop improvement, which cannot be obtained by exchange. The spread of improved varieties has resulted in the loss of indigenous crop genetic diversity. Whereas, industrialization and urbanization has seriously damaged the natural environment of crop plants. Human population is rapidly eroding the reservoir of genetic diversity. Habitats are disappearing at an alarming rate as forests are cleared, roads and cities expand, grasslands are plowed, burned and overgrazed, land is inundated by lakes impounded by new dams and new lands are irrigated for increased production of modern cultivars. The face of the earth is changing at an accelerating and alarming rate; and as it changes, more genetic diversity is lost forever. The Plant Exploration Laboratory has organized more than hundred expeditions in different agro-ecological regions of Pakistan to collect the targeted plant species. The main emphasis is to collect the major crops and their wild relatives as these species are under threat

Gene flow 5

Gene Flow:

“The introduction of genetic material (by interbreeding) from one population of a species to another”

Vertical Gene transfer: “The transfer of genes from parents to offspring.”

Horizontal gene transfer:

“Horizontal gene transfer is known to occur between different species, such as between prokaryotes and eukaryotes, between the three DNA-containing organelles of eukaryotes, the nucleus, the mitochondrion and the chloroplast.

Steps for conservation of plant genetic resources 5

The Global Strategy for Plant Conservation

The **Global Strategy for Plant Conservation (GSPC)** is a program of the UN's Convention on

Biological Diversity founded in 1999. It is a Plan to Save the World's Plant Species - grew out of the Convention on Biological Diversity and is being fed into government policy around the world.

Vision of GSPC:-

“Without plants, there is no life. The functioning of the planet, and our survival, depends on plants. The Strategy seeks to halt the continuing loss of plant diversity”

Objectives:

The GSPC highlights the importance of plants and the ecosystem services they provide for all life on earth, and aims to ensure their conservation. The GSPC has 5 main objectives:

1. Plant diversity is well understood, documented and recognized
2. Plant diversity is urgently and effectively conserved
3. Plant diversity is used in a sustainable and equitable manner
4. Plant diversity is used in a sustainable and equitable manner
5. The capacities and public engagement necessary to implement the strategy have been developed.

Critical Endangered Species.

A **critically endangered** (CR) species is one which has been categorized by the International Union for Conservation of Nature (IUCN) as facing an extremely high risk of extinction in the wild.

Examples:

Eastern Lowland Gorilla

Hawksbill Turtle

Javan Rhino